

26 explore and explain what is meant by the terms *brittle*, *ductile*, *hard*, *malleable*, *stiff* and *tough*. Use these terms, give examples of materials exhibiting such properties and explain how these properties are used in a variety of applications, for example, safety clothing, foodstuffs

7.5 Spare part surgery (SUR)

A study of the physics associated with spare part surgery for joint replacements and lens implants:

- mechanical properties of bone and replacement materials
- 'designer' materials for medical uses.

There are opportunities for students to consider ethical issues relating to surgical intervention, and to learn how new scientific knowledge is validated and communicated through peer-reviewed publication.

Students will be assessed on their ability to:	Suggested experiments
22 obtain and draw force–extension, force–compression, and tensile/compressive stress-strain graphs. Identify the <i>limit of proportionality</i> , <i>elastic limit</i> and <i>yield point</i>	Obtain graphs for, for example, copper wire, nylon and rubber
23 investigate, and use Hooke's law, $F = k\Delta x$, and know that it applies only to some materials	
24 explain the meaning of, use and calculate <i>tensile/compressive stress</i> , <i>tensile/compressive strain</i> , <i>strength</i> , <i>breaking stress</i> , <i>stiffness</i> and <i>Young Modulus</i> . Obtain the Young modulus for a material	Investigations could include, for example, copper and rubber
27 calculate the elastic strain energy E_{el} in a deformed material sample, using the expression $E_{el} = \frac{1}{2} Fx$, and from the area under its force/extension graph	